

Waseem Raza, PhD

Machine Learning Engineer | Financial AI/ML & Data Science

Financial ML | Forecasting | Risk Analytics | Data Engineering

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Summary

Machine Learning Engineer and Data Scientist with a PhD in Electrical and Computer Engineering, production ML experience, and finance-focused portfolio work in credit-risk modeling, loan-default prediction, time-series forecasting, and reinforcement-learning applications. Skilled in Python, SQL, scikit-learn, XGBoost, TensorFlow, PyTorch, Pandas, NumPy, statistical modeling, cross-validation, feature engineering, Azure Databricks, Apache Spark, ETL/ELT, predictive analytics, and model evaluation. Developing deeper specialization in financial forecasting, risk analytics, portfolio optimization, and data-driven decision intelligence.

Technical Skills

Financial Machine Learning: Credit-risk modeling, loan-default prediction, supervised learning, predictive modeling, reinforcement-learning concepts, portfolio and trading applications

Time Series & Statistics: Forecasting, statistical modeling, feature engineering, regression, classification, model diagnostics, validation, and uncertainty-aware analysis

Data Science & Analytics: Python, SQL, Pandas, NumPy, SciPy, Statsmodels, exploratory analysis, visualization, and data-quality assessment

Machine Learning: scikit-learn, XGBoost, TensorFlow, PyTorch, cross-validation, hyperparameter tuning, anomaly detection, and model evaluation

Data Engineering & Cloud: ETL/ELT, Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Synapse Analytics, BigQuery, and AI-ready datasets

MLOps & Engineering: Git, Docker, CI/CD, APIs, testing, reproducible workflows, model deployment, and production-oriented engineering

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

02/2025 - Present

Bedminster, NJ

- Lead end-to-end development, validation, and operational integration of machine-learning diagnostics, including data preparation, classification, model evaluation, and error analysis.
- Design domain-adaptive neural models and reusable data-processing pipelines that improve robustness across heterogeneous operational datasets and distribution shifts.
- Build cross-domain validation, data-quality, and failure-mode assessment workflows to support reliable predictive analytics and decision-support systems.
- Translate complex operational requirements into measurable ML objectives, validated datasets, and production-oriented analytical workflows.

Data Scientist and AI/ML Engineer

Impac Exploration Services

09/2024 - 12/2024

Weatherford, OK

- Developed scalable analytical and ETL pipelines using Python, Azure Databricks, Azure Functions, and Synapse Analytics for large structured datasets.
- Automated ingestion, cleansing, transformation, and validation workflows, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Translated domain requirements into AI-ready datasets and reproducible analytical workflows for data-driven decision support.

Graduate Research Assistant

AI4Networks, University of Oklahoma

09/2019 - 07/2025

Tulsa, OK

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.

- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering University of Oklahoma, Norman, Oklahoma, USA	09/2019 - 07/2025
MSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	08/2014 - 08/2016
BSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	11/2010 - 06/2014

Selected Projects

Financial ML and Forecasting <i>Portfolio Project</i> • Credit-risk and loan-default modeling alongside time-series forecasting workflows for financial data. <i>Tools & methods:</i> Python, Machine Learning, Time Series, Model Evaluation	2024-2025
AI-Ready Lakehouse Pipeline <i>Portfolio Project</i> • An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads. <i>Tools & methods:</i> Azure, Databricks, Spark, Data Engineering	2024
System Design Case-Study Portfolio <i>Guided Project</i> • Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs. <i>Tools & methods:</i> System Design, APIs, Distributed Systems, Scalability	2026 In progress

Certifications

1. Machine Learning and Reinforcement Learning in Finance Specialization — [New York University / Coursera](#) | Completed (0/4) courses

Focus: An applied machine-learning specialization focused on financial prediction, portfolio management, trading, risk modeling, market dynamics, and reinforcement-learning methods for finance.

Tools & Skills: Machine Learning for Finance, Reinforcement Learning, Financial Modeling, Portfolio Management, Financial Trading, Risk Modeling, Predictive Modeling, Supervised Learning, Unsupervised Learning, Statistical Machine Learning, Market Dynamics, Derivatives, TensorFlow, Scikit-learn, Python
2. Machine Learning — Stanford University / Coursera | Completed

Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization
3. Google Cloud Data Analytics Professional Certificate — Google Cloud / Coursera | Completed

Focus: Cloud data transformation, analysis, visualization, and decision support using Google Cloud tools.

Tools & Skills: SQL, BigQuery, Data Analytics, Visualization